

WHAT ARE TENSORS:

In the simplest form tensors are data structures, a container for numbers.

Why study tensor??

↳ All leading libraries make use of tensors.

Tensors form the most fundamental part of ML & DL.

0 D Tensor or scalar

↳ Tensor with no dimension, like (2).

0D means no dimension.

An n-dimensional array is a tensor.

1 D tensor / Vector

↳ a list of numbers: $[1, 2, 3, 4]$ → this is a vector of 4 dimensions

no. of dim. = 1 ✓
1-D

* Each tensor is a vector & the no. of observations in the vector = dimensions of the vector.

For eg: `arr = np.array([1, 2, 3, 4])`

is a 1D tensor but the dimension of the vector are = 4.

2-D tensor / Matrices

Matrices: collection of vectors.

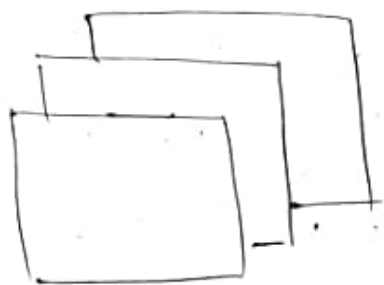
$[1, 2, 3]$ $[4, 5, 6]$ $[7, 8, 9]$

↳ combining we get a matrix:

$$\begin{bmatrix} [1, 2, 3] \\ [4, 5, 6] \\ [7, 8, 9] \end{bmatrix}$$

no. of dim = 2 (1 row dim & 1 column dim).

N-D tensor



A $4 \times 3 \times 3$ tensor

with 3 dimensions:

the row, the column & depth of matrices.

↳ collection of matrices.

A 4-D tensor would be a vector of 3-D tensors &

A 5-D tensor would be matrix of 3-D tensors
and so on...

In machine learning most problems revolve around
using tensors from 0D to 5-D. Higher dimensions are
very-very rarely required.

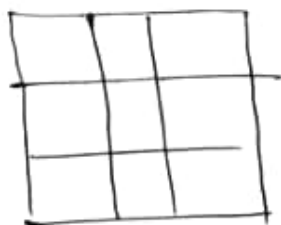
Rank, Axes & Shape

Simply put:

no. of axes = Rank = no. of dimensions.

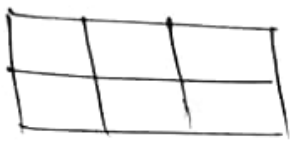
① Shape: No. of items that are present in a particular
axis.

eg:



n. dim = 2 (row & column)

but shape = (3, 3)



→ now dimensions for this also = 2
(row & column)
but shape here would be = $(3, 2)$

as we can store 3 values in a row & 2 in a column.

(*) Size of a tensor : no. of items that can be stored in the tensor.

↳ simply multiply the numbers in it's dimensions.

for eg: if shape of a tensor = $(3, 2)$

it's size would be = $3 \times 2 = 6$ — .

① Size of a scalar = $\boxed{1}$ ✓